

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 2

Academic Year: 2024

Subject: 031001122 Chemistry II

Section: 15-18

Date: 7 January 2025

Time: 13.00 – 15.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 parts. There will be 9 pages (including this page) and they are worth 40 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators are **NOT** allowed in the examination.
- Periodic table is attached on the last page of the midterm examination.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

		Exam
CLO 1	Gain an understanding of state of matters.	Parts 1-2
CLO 2	Comprehend solid, liquid and gas characteristics.	Parts 1-2
CLO 3	Learn gas's laws including terminologies related to solid, liquid and gas.	Parts 1-2
CLO 4	Solve chemistry problems and demonstrate reasoning clearly and completely for solid, liquid and gas.	Parts 1-2
CLO 5	Understand solution, solution preparation and solution dilution.	Parts 1-2
CLO 6	Solve problems related to solution, solution preparation and dilution clearly and rationally.	Parts 1-2

Instructor: Assoc. Prof. Dr. Nathjanan Jongkon and Asst. Prof. Dr. Sunanta Chuayprakong

	Part 1	Part 2	Total
score			

Gas constant or $R = 0.082 \text{ atm}\cdot\text{L/mol}\cdot\text{K}$

Part 1. Multiple Choice (5 points)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

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1. What is the primary factor that determines the state of matter of a substance?
a. Mass b. Volume c. Temperature d. Density
2. At high temperatures the particles of a gas
a. move slower and hit the walls of their container less often.
b. move faster and hit the walls of their container more frequently.
c. move faster and hit the walls of their container less frequently.
d. don't move at all and don't hit the walls of their container.
3. What is the effect of high temperature on surface tension?
a. It increases surface tension. b. It decreases surface tension.
c. It has no effect. d. It doubles surface tension
4. Which of the following best describes a crystalline solid?
a. A solid with a well-defined pattern of particles. b. A solid with no fixed shape.
c. A solid with randomly arranged particles. d. A solid that easily deforms under pressure
5. If a gas with an unknown molar mass diffuses at a rate twice as fast as oxygen gas, what is the estimated molar mass of the unknown gas?
a. 8 g/mol b. 16 g/mol c. 16 g/mol d. 64 g/mol
6. A container holds 2.5 moles of nitrogen gas at a temperature of 35°C and a pressure of 1.2 atm. What is the density of the nitrogen gas in g/L?"
a. 0.85 g/L b. 1.33 g/L c. 2.50 g/L d. 3.75 g/L
7. Which of the following solutions has the highest solution concentration?"
a. 15 grams of solute in 50 mL of water b. 25 grams of solute in 125 mL of water
c. 30 grams of solute in 150 mL of water d. 10 grams of solute in 25 mL of water
8. You need to prepare 500 mL of a 0.1 M NaCl solution by diluting a 1 M stock solution. How many milliliters of the stock solution do you need to use?
a. 2 b. 5 c. 20 d. 50
9. Which of the following statements is incorrect?
a. 6.02×10^{23} atoms for nitrogen gas and argon gas occupy the same volume at STP.
b. Sublimation is the opposite of deposition.
c. Gas particles move faster than particles in solids or liquids.
d. Most metals are in the solid state.
10. A mixture of CO₂ and an unknown gas could effuse from a container. CO₂ took 1.25 times as long to escape as the unknown gas. Which one could be the unknown gas?
a. Cl₂ b. H₂ c. CO d. HCl

Part 2 Read the construction carefully and complete the required information.

Students must provide calculation detail if CALCULATION is noted for any questions.

Not following these directions could affect your score. (35 points)

2.1 Read the following sentences. Write "T" if the sentence is correct and "F" if it is not.

(2.5 points)

- a. Dilution is related to adding more solvent and more solute.
- b. Effusion describes the movement of gas molecules through a very small opening due to a pressure difference.
- c. Weak intermolecular forces result in lower vapor pressure.
- d. An amorphous solid is a solid that lacks an ordered internal structure.
- e. One mole of any gas at STP occupies 22.4 L.

2.2 Consider the following situations below and complete them by writing **less, weaker, higher, stronger, or greater** in the blanks. (2.5 points)

- a. Solid molecules generally have intermolecular forces than liquid molecules.
- b. During diffusion, gas particles spread out from an area of concentration until the concentration becomes uniform throughout the space.
- c. Interaction in an amorphous solid is generally than that in a crystalline solid.
- d. A dilute solution has a concentration compared to a stock solution.
- e. According to Boyle's Law, when the pressure of a gas is high, the volume of the gas will be

2.3 Explain dynamic equilibrium vapor pressure when a liquid is enclosed in a chamber.
(2 points)

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2.4 Illustrate how particles are arranged in both amorphous and crystalline solids.

Hint: Utilize a black dot (•) to symbolize a solid particle." (2 points)

Amorphous solid	Crystalline solid

2.5 Draw particle diagrams to show the ice, the water and the stream of H_2O molecules and answer the questions below. (2 points)

Ice	Water	Stream

Which state does H_2O have the weakest molecular interaction? Why?

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.....

.....

2.6 A balloon had a volume of 8 L at 87°C and a pressure of 760 torr. Afterwards, its volume become a half at constant pressure. Calculate what temperature in Celsius must the balloon be cooled to become a half of the original volume. (CALCULATION, 2 points)

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- 2.7 Consider a 1.2 L sample of gas. What was its initial pressure when its volume is compressed to 1.6 L at a new pressure of 3.6 atm? (CALCULATION, 1 point)
- 2.8 At 47 °C and 140 kPa, sulfur dioxide gas occupies a volume of 40 dm³. Determine the final temperature of the gas if the volume is decreased to 20 dm³ at constant pressure. (CALCULATION, 2 point)
- 2.9 What is the volume of a container if it holds 0.2 g of neon gas at STP? (CALCULATION, 2 point)
- 2.10 A flask contains a mixture of Ar and Ne at a total pressure of 7.5 atm. There are 2.0 mol of Ar and 4.0 mol of Ne in the flask. Calculate the partial pressure of Ar. (CALCULATION, 2 point)

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2.11 Diatomic molecules 4.63 g of a gas occupy 2.73 L at STP. Can you identify the gas?

Hint: Look at the periodic table for the gas identification. (CALCULATION, 2 points)

2.12 The reaction of calcium carbonate (CaCO_3) and hydrochloric acid (HCl) produces calcium chloride (CaCl_2), water (H_2O), and carbon dioxide gas (CO_2). If the reaction takes place in a container at 0.82 atm and 27.0 °C. How many liters of CO_2 are released when 50.0 g of CaCO_3 reacts with enough HCl? (CALCULATION, 3 points)



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2.13 Compare effusion rate between oxygen gas and helium gas. Based on your calculated effusion rates, indicate oxygen gas or helium gas can effuse slower.

(CALCULATION, 2 points) Hint: $\sqrt{2} = 1.414$ and $\sqrt{3} = 1.732$

2.14 Calculate molarity, molality, and mole fraction of 10% w/w of CaCO_3 . The density of solution is 2.5 g/cm^3 (CALCULATION, 3 points)

a. molarity

b. molality

c. mole fraction

2.15 Calculate a molar concentration in each of the following solutions:

(CALCULATION, 3 points)

a. 48 g of lithium hydroxide in 200 mL of solution

b. 6.02×10^{22} calcium bromide (CaBr_2) molecules in 25 mL of solution

c. According to the molar concentrations as calculated above, which one is more concentrated?

2.16 What volume of water would you add to 15.00 mL of a 6.00 M solution of nitric acid (HNO_3) in order to get a 1.50 M solution? (CALCULATION, 2 points)

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Periodic Table of the Elements

1 hydrogen 1 H 1.0079	2 helium 2 He 4.0026																
3 lithium 3 Li 6.941	4 beryllium 4 Be 9.0122																
11 sodium 11 Na 22.990	12 magnesium 12 Mg 24.305															13 boron 5 B 10.811	14 carbon 6 C 12.011
19 potassium 19 K 39.098	20 calcium 20 Ca 40.078															15 nitrogen 7 N 14.007	16 oxygen 8 O 15.999
37 rubidium 37 Rb 85.468	38 strontium 38 Sr 87.62															17 fluorine 9 F 18.998	18 neon 10 Ne 20.180
55 caesium 55 Cs 132.91	56 barium 56 Ba 137.33															19 argon 18 Ar 39.948	
87 francium [223]	88 radium [226]	57-70 ■	3 scandium 21 Sc 44.956	4 titanium 22 Ti 47.867	5 vanadium 23 V 50.942	6 chromium 24 Cr 51.996	7 manganese 25 Mn 54.938	8 iron 26 Fe 55.845	9 cobalt 27 Co 58.933	10 nickel 28 Ni 58.693	11 copper 29 Cu 63.546	12 zinc 30 Zn 65.39	13 aluminum 13 Al 26.982	14 silicon 14 Si 28.086	15 phosphorus 15 P 30.974	16 sulfur 16 S 32.065	17 chlorine 17 Cl 35.453
			19 yttrium 39 Y 88.906	20 zirconium 40 Zr 91.224	21 niobium 41 Nb 92.906	22 molybdenum 42 Mo 95.94	23 technetium 43 Tc [98]	24 ruthenium 44 Ru 101.07	25 rhodium 45 Rh 102.91	26 palladium 46 Pd 106.42	27 silver 47 Ag 107.87	28 cadmium 48 Cd 112.41	31 gallium 31 Ga 69.723	32 germanium 32 Ge 72.61	33 arsenic 33 As 74.922	34 selenium 34 Se 78.96	35 bromine 35 Br 79.904
			71 lutetium 71 Lu 174.97	72 hafnium 72 Hf 178.49	73 tantalum 73 Ta 180.95	74 tungsten 74 W 183.84	75 rhenium 75 Re 186.21	76 osmium 76 Os 190.23	77 iridium 77 Ir 192.22	78 platinum 78 Pt 195.08	79 gold 79 Au 196.97	80 mercury 80 Hg 200.59	49 indium 49 In 114.82	50 tin 50 Sn 118.71	51 antimony 51 Sb 121.76	52 tellurium 52 Te 127.60	53 iodine 53 I 126.90
			103 lawrencium [262]	104 rutherfordium [261]	105 dubnium [262]	106 seaborgium [266]	107 bohrium [264]	108 hassium [269]	109 meitnerium [268]	110 darmstadtium [271]	111 roentgenium [272]	112 ununbium [277]	81 thallium 81 Tl 204.38	82 lead 82 Pb 207.2	83 bismuth 83 Bi 208.98	84 polonium [209]	85 astatine [210]
		89-102 ■	118 ununoctium [294]	117 ununseptium [291]	116 ununhexium [292]	115 ununpentium [288]	114 ununquadium [289]	113 ununtrium [284]	112 ununbium [277]	111 roentgenium [272]	110 darmstadtium [271]	109 meitnerium [268]	108 hassium [269]	107 bohrium [264]	106 seaborgium [266]	105 dubnium [262]	104 rutherfordium [261]

■ Lanthanide series

■ Actinide series

lanthanum 57 La 138.91	cerium 58 Ce 140.12	praseodymium 59 Pr 140.91	neodymium 60 Nd 144.24	promethium 61 Pm [145]	samarium 62 Sm 150.36	europium 63 Eu 151.96	gadolinium 64 Gd 157.25	terbium 65 Tb 158.93	dysprosium 66 Dy 162.50	holmium 67 Ho 164.93	erbium 68 Er 167.26	thulium 69 Tm 168.93	ytterbium 70 Yb 173.04
actinium 89 Ac [227]	thorium 90 Th 232.04	protactinium 91 Pa 231.04	uranium 92 U 238.03	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	einsteinium 99 Es [252]	fermium 100 Fm [257]	mendelevium 101 Md [258]	nobelium 102 No [259]